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UNIT 3: CREATIVE INVENTIONS

Chapter 19: The Father of Playstation
(Ken Kutaragi) & Video Gaming Dynamics

LIVRET D'EXCELLENCE — PROGRAMME TUNISIEN DE TERMINALE (BAC)

Target Audience: Year 4 Secondary Education (Baccalauréat)

Unit Focus: Creative Inventions & Technological Pioneers

Linguistic Elements: Verb Patterns (Gerunds/Infinitives) & Opinion Exponents

Phonetics Focus: Word Stress, Complex IPA & Past Tense Endings

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PART 1: THE EXHAUSTIVE COURSE THEORY

1.1 Biographical Portrait: Ken Kutaragi & The PlayStation Revolution

Ken Kutaragi (born August 8, 1950) is widely known as the 'Father of the PlayStation'. As a brilliant hardware engineer at Sony, he observed his daughter playing with a Nintendo Famicom and realized the massive untapped potential of dedicated home gaming consoles. Despite immense resistance from senior Sony executives who viewed video games as a passing fad or cheap toys beneath Sony's corporate dignity, Kutaragi persisted with exceptional drive and stubbornness. He secretly developed a state-of-the-art digital sound chip (the SPC700) for Nintendo's Super NES. When discovered, it almost cost him his job, but Sony's CEO Norio Ohga recognized his genius and backed his project.

Kutaragi's ultimate breakthrough was the creation of the original Sony PlayStation (released in 1994), which featured unprecedented 3D polygon graphics processing chips. His engineering prowess revolutionized the gaming industry, pushing it away from 2D cartridge games to high-capacity CD-ROMs. Through his relentless efforts, Sony went on to dominate the global entertainment market, transforming gaming from a subculture into a multi-billion dollar mainstream industry.

1.2 Core Vocabulary Portfolio (Bac Essential)

Word & IPA	POS	Academic Meaning & Synonyms	Contextual Example
stardom /'stɑːdəm/	Noun	The state of being a famous performer; celebrity status, renown.	His invention catapulted him to global stardom.
prowess /'praʊ.əs/	Noun	Extraordinary skill, expertise, or bravery in a particular activity.	Kutaragi's engineering prowess was unmatched.
convene /kən'viːn/	Verb	To assemble or bring together for a formal meeting or activity.	Sony executives convened to debate the game console project.
stubbornness /'stʌb.ən.nəs/	Noun	Determination not to change one's attitude or opinion on something.	His stubbornness kept the PlayStation project alive.
faint-hearted /feɪnt'haː.tɪd/	Adj	Lacking courage; timid, cowardly. Not for the weak.	Pioneering new technologies is not for the faint-hearted.
to pioneer /ˌpaɪə'nɪər/	Verb	To develop or be the first to use a new method, area, or activity.	Sony pioneered the widespread use of 3D polygon graphics.
breakthrough /'breɪk.θruː/	Noun	A significant and dramatic development or obstacle clearance.	The 3D graphics chip was a major technological breakthrough.

1.3 Advanced Grammar: Verb Patterns (Gerunds & Infinitives)

A critical component of the Baccalauréat grammar syllabus involves mastering verb complementation patterns, specifically understanding which structures require a Gerund (V-ing) versus an Infinitive (to + Verb).

Rule A: Verbs Followed Strictly by the Gerund (-ing):

Verbs of feeling (dis)likes, continuation, and specific idioms require the gerund form:

- Enjoy / Mind / Avoid / Postpone / Risk / Admit / Deny + V-ing
- Keep / Keep on / Carry on + V-ing
- Prepositional Idioms (Exam Alert!): Look forward to / Be keen on / Be devoted to / Be dedicated to / Object to / Can't stand / It's no use + V-ing.

Example: Kutaragi kept on working (not 'to work') on the polygon chip. He looked forward to proving them wrong.

Rule B: Verbs Followed Strictly by the Infinitive (to + Verb):

Verbs of intent, plan, attempt, and choice require the infinitive form:

- Decide / Plan / Hope / Aim / Agree / Attempt / Offer / Promise / Refuse / Manage + To-Verb.

Example: Sony executives managed to secure global market dominance through innovation.

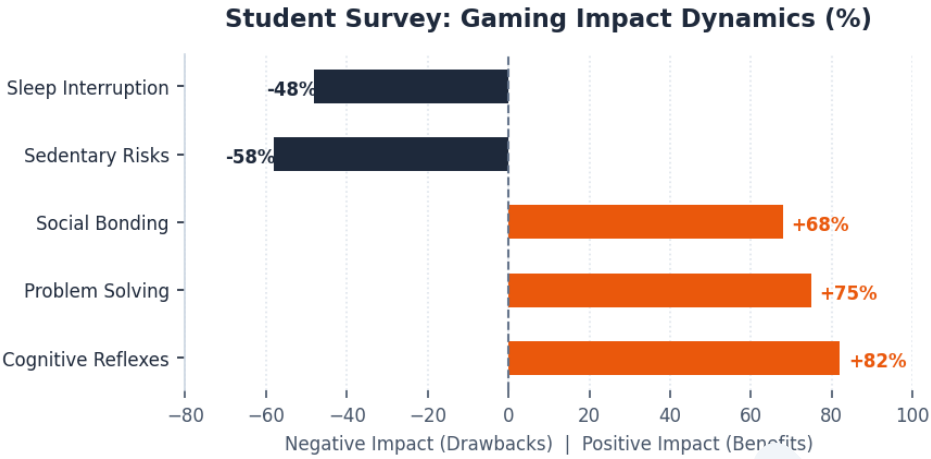
1.4 Phonetics Module: Stress Patterns & Consonant Clusters Discrimination

In the Bac exam, phonetics questions focus on identifying primary word stress shifts and pronouncing complex consonant clusters.

- Word Stress Shifts in Word Families: Notice how the primary stress moves depending on the suffix:
 - IN-vent /'ɪn.vent/ (Verb - 2 syllables, stress on 1st) ↔ in-VEN-tion /ɪn'ven.ʃən/ (Noun - 3 syllables, stress on 2nd).
 - STARD-om /'stɑ:dəm/ (Noun - 2 syllables, stress on 1st) ↔ grow-ESS /'grəʊ.əs/ (Noun - 2 syllables, stress on 1st).
 - per-se-VERE /,pɜ:.sɪ'vɪər/ (Verb - stress on 3rd) ↔ per-se-VER-ance /,pɜ:.sɪ'vɪə.rəns/ (Noun - stress on 3rd).
- Pronunciation of '-ed' Endings (Past Simple):
 - `/ɪd/` after /t/ or /d/: insisted, devoted, dedicated, accelerated.
 - `/t/` after voiceless sounds (/p/, /k/, /f/, /tʃ/, /s/, /ʃ/): pioneered (/d/), watched (/t/), risk (/t/).
 - `/d/` after voiced sounds: revolutionized, resolved, backed (/t/).

1.5 Data Analysis: Youth Perspectives on Video Gaming

According to survey data featured in Unit 3, secondary education students experience both intellectual enrichment and potential sedentary lifestyle risks. The chart below synthesizes these conflicting impact rates.



PART 2: THE ACTIVE RECALL REPOSITORY (40 FLASHCARDS)

Use this double-column table for active recall. Cover the right-hand column and test yourself on the questions below.

Exam Question / Recall Trigger	Detailed Model Answer & Exam Key
1. Q1: Who is Ken Kutaragi?	He is a brilliant Sony hardware engineer universally known as the 'Father of the PlayStation' for inventing and developing the PlayStation console.
2. Q2: What motivated Kutaragi to develop a gaming console?	He watched his young daughter playing with a Nintendo Famicom and realized the massive potential of dedicated home video game systems.
3. Q3: What internal obstacle did Kutaragi face at Sony?	Senior executives strongly opposed the project, viewing video games as cheap toys or passing fads beneath Sony's corporate identity.
4. Q4: What was Kutaragi's first secret breakthrough project?	He secretly developed the high-quality digital sound chip SPC700 for Nintendo's Super NES console while still employed at Sony.
5. Q5: Who rescued Kutaragi's career when his secret work was discovered?	Sony's visionary CEO Norio Ohga recognized Kutaragi's engineering genius and fully backed his gaming project.
6. Q6: What made the original PlayStation technically revolutionary in 1994?	It pioneered the widespread use of high-capacity CD-ROMs instead of cartridges and introduced advanced 3D polygon graphics rendering.
7. Q7: Define 'stardom' as used in Chapter 19 context.	'Stardom' means reaching celebrity status or supreme fame. Kutaragi achieved global stardom after the release of the PlayStation.
8. Q8: Define the word 'prowess'.	Prowess refers to extraordinary skill, expertise, or exceptional ability in a specific field (e.g., Kutaragi's technical prowess).
9. Q9: What does the term 'convene' mean?	To convene means to assemble or call together a group of people for an official meeting or structured discussion.
10. Q10: What does 'stubbornness' signify in Kutaragi's biography?	It represents his relentless determination and refusal to abandon his vision, even when facing extreme corporate discouragement.
11. Q11: Translate 'faint-hearted' and provide a synonym.	Faint-hearted means lacking courage or being timid. Synonyms include fearful, cowardly, or weak-willed.
12. Q12: What does 'to pioneer' mean as a verb?	It means to be the first to develop, introduce, or utilize a new technology, method, or field of study.
13. Q13: Define 'breakthrough' with an example.	A breakthrough is a significant technological or scientific development that overcomes an obstacle. (e.g., Kutaragi's 3D chip).

14. Q14: Why was CD-ROM technology preferred over traditional cartridges?	CD-ROMs offered vastly superior storage capacity at a much lower manufacturing cost, allowing richer audio and expansive 3D games.
15. Q15: How did Kutaragi's stubbornness pay off eventually?	His stubbornness led to the launch of the PlayStation, which dominated the global entertainment market and generated immense profit for Sony.
16. Q16: Which verb pattern follows 'look forward to'?	It is followed strictly by a Gerund (V-ing) because 'to' is a preposition here. Example: 'He looks forward to releasing the new console.'
17. Q17: Is 'enjoy' followed by a gerund or an infinitive?	Enjoy is strictly followed by a Gerund (V-ing). Example: 'Millions of teenagers enjoy playing PlayStation games.'
18. Q18: Correct the error: 'He managed surviving the harsh executive meeting.'	Correction: 'He managed to survive the harsh executive meeting.' (Manage requires the infinitive 'to + verb').
19. Q19: What is the rule for verbs like 'avoid', 'postpone', and 'risk'?	These verbs must be followed strictly by the Gerund (V-ing). Example: 'We must avoid delaying the product launch.'
20. Q20: Formulate a sentence with 'be keen on' using Chapter 19 vocabulary.	Example: 'Young gamers are keen on buying state-of-the-art interactive devices.' ('keen on' + V-ing).
21. Q21: Which verb pattern follows 'decide', 'plan', and 'agree'?	They are followed strictly by the Infinitive (to + verb). Example: 'Sony decided to invest millions in new hardware.'
22. Q22: Explain the grammar rule of 'be devoted to'.	Since 'to' is a preposition in this idiom, it must be followed by a Gerund (V-ing). Example: 'He was devoted to designing microchips.'
23. Q23: Correct this sentence: 'They postponed to launch the PlayStation 2.'	Correction: 'They postponed launching the PlayStation 2.' (Postpone requires V-ing).
24. Q24: What is the grammar pattern for the idiom 'can't stand'?	It is followed by a Gerund (V-ing). Example: 'Kutaragi could not stand wasting time in pointless executive debates.'
25. Q25: What verb pattern follows 'attempt' and 'fail'?	Both verbs require an Infinitive (to + verb). Example: 'Sony executives attempted to cancel Kutaragi's project.'
26. Q26: Rewrite using 'unless': 'If we do not innovate, we will lose market share.'	Unless we innovate, we will lose market share. (Unless replaces 'if... not' and takes an affirmative verb).
27. Q27: Provide the correct pattern: 'It is no use (argue) with him.'	It is no use arguing with him. ('It is no use' is a prepositional idiom requiring V-ing).
28. Q28: Which pattern follows 'promise' and 'refuse'?	They are followed by the Infinitive (to + verb). Example: 'CEO Norio Ohga promised to support Kutaragi's research.'

29. Q29: What is the difference between 'stop doing' and 'stop to do'?	'Stop doing' means ceasing an action entirely. 'Stop to do' means pausing one action in order to perform another. Example: 'He stopped working on sound chips' vs 'He stopped to drink some water.'
30. Q30: Complete correctly: 'He is dedicated to (pioneer) interactive tech.'	He is dedicated to pioneering interactive tech. ('dedicated to' takes a Gerund).
31. Q31: What is the primary word stress of the verb 'invent'?	The primary stress is on the second syllable: in-VENT /In'vent/.
32. Q32: Where does the stress fall on the noun 'invention'?	The primary stress is on the second syllable, right before the '-tion' suffix: in-VEN-tion /In'ven.fən/.
33. Q33: Identify the silent letters in the word 'knight' and 'castle'.	In 'knight', the 'k' and 'gh' are silent. In 'castle', the 't' is silent.
34. Q34: How is the past tense ending '-ed' pronounced in 'insisted'?	It is pronounced as /ɪd/ because the root verb ends with the alveolar consonant /t/.
35. Q35: Classify the '-ed' ending pronunciation of 'watched'.	It is pronounced as /t/ because the root verb ends with the voiceless sound /t/.
36. Q36: Pronounce 'perseverance' and locate its primary stress.	Pronunciation: /ˌpɜːsɪvɪə.rəns/. The primary stress is on the third syllable: per-se-VER-ance.
37. Q37: What are the primary benefits of video gaming according to the survey?	The main benefits include improved cognitive reflexes, rapid problem-solving skills, and enhanced social bonding in multiplayer modes.
38. Q38: What are the primary drawbacks of excessive gaming?	Excessive gaming can cause severe sleep interruption, screen addiction, and sedentary health risks.
39. Q39: What is the correct pronunciation of Kutaragi's invention 'PlayStation'?	Pronunciation: /ˈpleɪ.steɪ.ʃən/. The primary stress falls on the first syllable: PLAY-station.
40. Q40: How does 'Unless' change a conditional sentence?	It replaces 'If ... not' by turning the negative condition into an affirmative clause. There can be no double negation (no 'not' after unless).

PART 3: EXAM EVALUATION QUIZ & CORRECTION

3.1 Diagnostic Mock Exam (10 Questions)

Answer the following mock exam questions to test your ultimate readiness. Questions 1-5 are Multiple Choice (MCQ). Questions 6-10 are True/False with mandatory justification.

1. Ken Kutaragi's sound chip SPC700 was secretly designed for which company?

A) Sega B) Nintendo C) Atari D) Commodore

2. Which of the following verbs is strictly followed by an Infinitive (to + verb)?

A) enjoy B) postpone C) attempt D) avoid

3. What is the correct pronunciation of the '-ed' ending in 'revolutionized'?

A) `/ɪd/` B) `/t/` C) `/d/` D) It is silent

4. Complete: 'He is extremely keen on _____ interactive software

A) develop B) developing C) to develop D) developed

5. Where is the primary word stress located in the noun 'perseverance'?

A) 1st syllable B) 2nd syllable C) 3rd syllable D) 4th syllable

6. [True/False] Sony executives enthusiastically supported Kutaragi's original PlayStation project from the beginning.

Justification: _____

7. [True/False] 'Look forward to' is followed by an infinitive with to.

Justification: _____

8. [True/False] In the word 'casual', the letter 't' is silent.

Justification: _____

9. [True/False] The survey chart indicates that cognitive reflexes are considered a major drawback of gaming.

Justification: _____

10. [True/False] Unless is grammatically equivalent to 'If ... not', and its following verb must be negative.

Justification: _____

3.2 Pedagogical Correction & Grading Key

Q.	Model Answer & Pedagogical Justification
Q1	B) Nintendo. Kutaragi secretly developed the SPC700 audio processing chip for Nintendo's Super NES, which almost led to his termination from Sony until Norio Ohga intervened.
Q2	C) attempt. Verbs of planning or attempting like 'attempt', 'decide', or 'manage' require 'to + verb' (e.g., 'They attempted to cancel the project').
Q3	C) <code>`/d/`</code> . Because the root verb 'revolutionize' ends in the voiced sound <code>/z/</code> , the past tense ending is pronounced as a voiced <code>`/d/`</code> .
Q4	B) developing. The idiom 'keen on' ends in a preposition, which strictly demands a Gerund (V-ing) as its object.
Q5	C) 3rd syllable. In 'perseverance' <code>/ˌpɜː.sɪˈvɪə.rəns/</code> , the primary accent falls on the third syllable: per-se-VER-ance.
Q6	FALSE. Senior Sony executives strongly opposed his idea, dismissing video games as children's toys and a waste of corporate resources.
Q7	FALSE. 'Look forward to' is a prepositional idiom. The 'to' here is a preposition, not an infinitive marker, so it must be followed by a Gerund (V-ing).
Q8	TRUE. In English phonology, the 't' in 'castle' <code>/ˈkɑːsl/</code> is a silent letter and is not pronounced.
Q9	FALSE. According to the data, 'cognitive reflexes' are a major benefit (+82%), not a negative drawback.
Q10	FALSE. While 'Unless' does replace 'If ... not', the clause following it must be in the affirmative form to avoid an ungrammatical double negation.

3.3 Master Revision Memo

Memo 1.1: Standard Verb Implementation Patterns (Gerunds vs. Infinitives)

Gerund (V-ing) Strictly	Infinitive (To + Verb) Strictly	Prepositional Idioms (+ V-ing)
enjoy, avoid, postpone, risk, admit, deny, keep, carry on	decide, plan, hope, aim, agree, attempt, refuse, manage, offer	look forward to, be keen on, be devoted to, object to, can't stand

Memo 1.2: Essential Interactive Writing Structure: Video Game Debate

Pros of Video Gaming	Cons of Video Gaming	Key Transition Words (Contrast)
Boosts spatial awareness & cognitive reflexes, teaches online collaboration, provides complex problem-solving scenarios.	Causes screen addiction, sedentarism leading to physical issues, and sleep disruption due to blue light.	whereas, while, unlike, in contrast, however, on the other hand, conversely